Machine Learning: Math4ml, Shallow Learning, and Deep Learning

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1 Machine Learning is Hot

It is hard to overstate the hype and glamour machine learning (and its superset AI) have received in the past years. Concretely, from Stanford HAI,

(i) Economic: "In 2021, global private investment in AI totaled around $93.5 billion, which is more than double the total private investment in 2020. That marks the greatest year-over-year increase since 2014 (when investment from 2013 to 2014 more than doubled)." [1]

(ii) Academic: "More scholars are focusing on pattern recognition and machine learning. Publications in these two areas have more than doubled since 2015. Other areas strongly influenced by deep learning, such as computer vision, data mining and natural language processing, have shown smaller increases." [1]

(iii) Government: "In the past few years, federal lawmakers have sharply increased proposed regulations involving AI. In 2015, just one federal bill was proposed, while in 2021, there were 130. Although this jump is significant, the number of bills related to AI being passed has not kept pace with the growing volume of proposed AI-related bills. This gap was most evident in 2021, when only 2% of all federal-level AI-related bills were ultimately passed into law." [1]

There is however a non-trivially sized community skeptical of the promises of AI. Within the AI debate, there are two major questions: (1) is AI hyped or overhyped and (2) is AI progress fruitful or is it deleterious?

Before joining in on this debate, we offer a summary of the history of machine learning, the problems it is good at solving – and conversely the problems where it is frail –, and the state of the art today.

1.1 History of Machine Learning

We highlight the most important events:

–

–

1.2 Tank to a Gun Fight

1.3 Problems where we need ML

1.4 State of the Art Today

1.5 Future of Machine Learning

2 Linear Algebra

Linear Algebra is the branch of mathematics concerning vector spaces, often finite or countable infinite dimensional, as well as linear mappings between such spaces. Such an investigation is initially motivated by a system of linear equations in several unknowns. [source]

Such equations are naturally represented using the formalism of matrices and vectors, since they compactly represent the linear equations, as well as make them easier to operate. For example, take the the following set of linear equations:

$$2x_1 + 2x_2 = 5x_3$$
$$7x_1 + x_2 = 2x_3$$

We can frame the following system with the following:

$$Ax = b$$

Where,

$$A = \begin{bmatrix} 2 & 2 \\ 7 & 1 \end{bmatrix}, b = \begin{bmatrix} 5 \\ 2 \end{bmatrix}$$

In this section we present important classes of spaces in which our data will live and our operations will take place: vector spaces, metric spaces, normed spaces, and inner product spaces. Generally speaking, these are defined in such a way as to capture one or more important properties of Euclidean space but in a more general way. [source]

2.1 Linear Equation

A **linear equation** is an equation in which the highest power of any variable is 1. A linear equation with one variable takes on the form: [source]

$$a_1x_1 + b = 0$$

More generally, a linear equation is any equation which can be expressed in the form: [source]

$$a_1x_1 + a_2x_2 + \ldots + a_nx_n + b = 0$$

Where $a_i \in \{1, n\}$ is any expression not containing the variable, often a real number.

2.1.1 Geometry of Linear Equations

More intuitively, we say it is a linear equation, because geometrically the equation strictly paints a line. For one variable, it paints a line in 2-dimensional space; two variables, paints a line in 3-dimensional space; and so forth.

An example of two linear equations which form a **system of linear equations**:

$$\begin{aligned} 2x - \ \ y &= 0 \\ x + 3y &= 7 \end{aligned}$$

We can view the two equations from two lenses. The first lens, observes the equations as **rows**: each equation individually. When we look at it in this light, we can deduce that the first equation has slope 2, y-intercept at 0, and x-intercept at 0; and the second equation has slope $\frac{1}{3}$, y-intercept at $\frac{3}{7}$ and x-intercept at 7. From there, one can determine the solution by seeing where the two lines intersect from their plots.

[insert plot]

The other lens in which we can view the system is from the **column** perspective. Here, we can rewrite the two equations as one **vector equation**.

$$x \begin{bmatrix} 2 \\ 1 \end{bmatrix} + y \begin{bmatrix} -1 \\ 3 \end{bmatrix} = \begin{bmatrix} 0 \\ 7 \end{bmatrix}$$

Here, we need to find the combination of vectors on the left side that produces the vector on the right side.

[insert plot]

n = 3

This brings us to a generlization: **row picture**: Intersection point of n planes **column picture**: The right side b is a combination of the column vectors **Solution to equations**: Intersection point of planes = coefficients in the combination of columns

2.2 Vector

A **vector** is a 1-D array of numbers or **scalars**. [source]

$$\vec{v} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

A vector can mean different things to different people, though. To a topologist, these arrays represent **euclidean vector** but for simplicity is called **vector**.

For example, [source] [source]

$$\vec{s} = \begin{bmatrix} 5 \\ 2 \end{bmatrix}$$

Has magnitude X and moves 5 units perpendicular to the first axis and 2 units perpendicular to the second axis points blank degrees from the origin.

To a machine learning engineer, a vector is particularly useful to represent data. For example, we can use a vector to represent the weight, height, and speed of dogs.

$$\vec{d} = \begin{bmatrix} 50 \\ 32 \\ 22 \end{bmatrix}$$

This particular dog, has a weight of 50kg, height of 32 inches, and speed of 22mph.

2.2.1 Vector Addition

Adding vectors $\vec{a}$ and $\vec{b}$ is quite natural; you simply sum them element-wise:

$$\begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix} + \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix} = \begin{bmatrix} a_1 + b_1 \\ a_2 + b_2 \\ \vdots \\ a_n + b_n \end{bmatrix}$$

[source]

[source]

2.2.2 Vector Subtraction

We can view vector subtraction from the lens of vector addition: Subtracting vector $\vec{b}$ from $\vec{a}$ (i.e. $\vec{a} - \vec{b}$), is the same as adding $\vec{a}$ with $-\vec{b}$ (i.e. $\vec{a} + (-\vec{b})$:

$$\begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix} + \begin{bmatrix} -b_1 \\ -b_2 \\ \vdots \\ -b_n \end{bmatrix} = \begin{bmatrix} a_1 + (-b_1) \\ a_2 + (-b_2) \\ \vdots \\ a_n + (-b_n) \end{bmatrix} = \begin{bmatrix} a_1 - b_1 \\ a_2 - b_2 \\ \vdots \\ a_n - b_n \end{bmatrix} = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix} - \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix}$$

[source]

2.2.3 Scalar-Vector Multiplication

2.2.4 Vector Multiplication (Dot Product)

The **dot product** or **scalar product** or **projection product** is the sum of the products of the corresponding entries of two equal length vectors:

$$\vec{a} \cdot \vec{b} = a_1 * b_1 + a_2 * b_2 + \ldots + a_n * b_n$$

Geometrically, the dot product can be viewed as the product of the two **magnitudes** or norms and the cosine of the angle between them:

$$\vec{a} \cdot \vec{b} = ||a||\ ||b||\ cos(\theta)$$

Alternatively, it is defined as the product of the projection of the first vector onto the second vector and the magnitude of the second vector

2.2.5 Vector Multiplication (Cross Product)

2.2.6 Linear Combinations of Vectors

A **linear combination** of vectors is defined by multiplying the vectors by constants over a field F and summing them together. For example, a linear combination of two vectors $\vec{v}$ and $\vec{u}$ is expressed as:

$$\alpha\vec{v} + \beta\vec{u} = \alpha \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} + \beta \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \begin{bmatrix} \alpha v_1 + \beta u_1 \\ \alpha v_2 + \beta u_2 \end{bmatrix}$$

for some scalars α, $\beta \in F$. [source]

For a set vectors $\vec{v_1}\vec{v_2}\ldots\vec{v_n}$ and scalars $\alpha_1\alpha_2\ldots\alpha_n$, we can express the linear combination compactly:

$$\sum_i^n \alpha_i\vec{v_i} := \alpha_1\vec{v_1} + \alpha_2\vec{v_2} + \ldots + \alpha_n\vec{v_n}$$

[source]

Additionally, if one vector equals the some of scalar multiples of other vectors, it is said to be a linear combination of the other vectors. [source]

We provide an example:

$$\vec{v} = \begin{bmatrix} 5 \\ 14 \end{bmatrix} \vec{u} = \begin{bmatrix} -2 \\ 1 \end{bmatrix} = \vec{w} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$$

Then for $\alpha = 2$, $\beta = 3$,

$$2 \begin{bmatrix} -2 \\ 1 \end{bmatrix} + 3 \begin{bmatrix} 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 5 \\ 14 \end{bmatrix}$$

So we say $\vec{v}$ is a linear combination of $\vec{u}$ and $\vec{w}$.

2.2.7 Linear Independence of Vectors

For $S = \{\vec{v_1},\ \vec{v_2},\ \ldots,\ \vec{v_n}\}$, we say the set of vectors S from a vector space V are **linearly dependent** iff there exists scalars α_1, α_2, $\ldots$, α_n <u>not all zero</u> such that:

$$\alpha_1\vec{v_1} + \alpha_2\vec{v_2} + \ldots + \alpha_n\vec{v_n} = \vec{0}$$

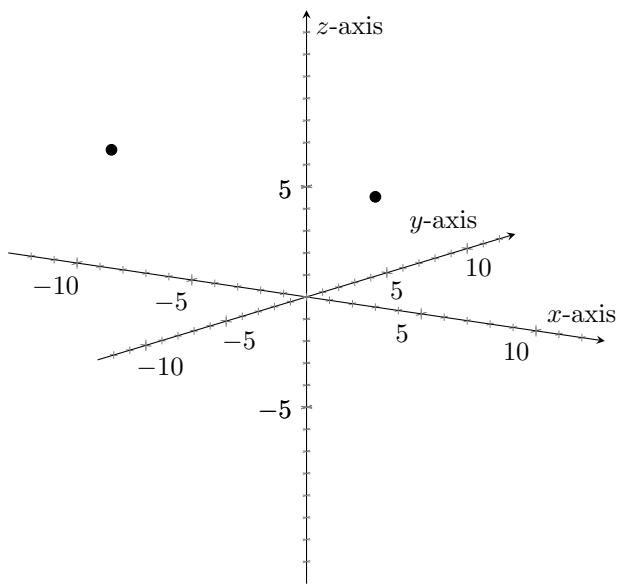
In words, the set of vectors are linearly dependent if there is a non-trivial linear combination that results in the zero vector. If no such linear combination exists,

the vectors are said to be **linearly independent**. More formally, they are linearly independent iff the above equation can only be satisfied by $\alpha_i = 0$ for $i = 1, \ldots, n$. [source]

The maximum number of linearly independenet vectors in a vector space equates to the **dimension** of the vector space. [source]

Cases:
(1) n = 1: $\vec{v}$ is linearly dependent of $\vec{u}$: Here, $\vec{v}$ and $\vec{u}$ point in the same direction on the same line.
(2) n = 2: $\vec{v}$ is linearly dependent of $\vec{u}$ and $\vec{w}$ (and $\vec{u} \neq \lambda\vec{w}$). Here $\vec{v}$ lies on the plane in $\mathbb{R}^j (j \geq 2)$ formed by $\vec{u}, \vec{w}$.
(3) n = 3: $\vec{v}$ is linearly dependent of $\vec{u}$, $\vec{w}$, $\vec{r}$ (and $\vec{u}, \vec{w}, \vec{r}$ are linearly independent). Here $\vec{v}$ lies in the sphere in $\mathbb{R}^j (j \geq 3)$ formed by $\vec{u}, \vec{w}$, $\vec{r}$.
(show one arrow l.i and one arrow l.d in each plot) [insert picture] [insert picture] [insert picture]

2.2.8 Vector Norm



A **norm** is a function that maps the real/complex vector space to the non-negative (greater than or equal to zero) reals and provides some notion of distance. [source]

The norm of x or $|x|$ or $||x||$ follows:

$$||x|| > 0 \text{ when } x \neq 0 \text{ and } ||x|| = 0 \text{ iff } x = 0$$
$$||k\,x|| = ||k||\,||x||$$
$$||x + y|| \leq ||x|| + ||y||$$

[source]

The **L^p-norm** or **p-norm** for $p = 1, 2, \ldots$ is defined as: [source]

$$|x|_p \equiv \left(\sum_i |x_i|^p \right)^{1/p}$$

For $p = 1$, we get the **taxicab norm**, where the distance defined for the **l^1 norm** is the manhattan distance:

$$|x|_1 = \sum_i x_i$$

For $p = 2$, the l^2 norm or **Euclidean norm**, often called norm, gives the distance from the origin:

$$|x|_2 = \sqrt{\sum_i x_i^2}$$

As $p \to \infty$ we get the **infinity-norm**:

$$|x|_\infty = \max_i |x_i|$$

[source]

2.2.9 Vector Projection

The **vector projection** of a on (or onto) a vector b is a vector parallel to b corresponding to how much of vector a goes in the direction of vector b: [code]

$$\text{proj}_{\mathbf{b}}\, \mathbf{a} = \frac{a \cdot b}{||b||} \frac{b}{||b||}$$

The **scalar projection** of a onto b is the amount, a_1, that a is in the direction of b.

$$a_1 = ||a||\, cos(\theta) = \, a \cdot \hat{b}$$

Notice that if they are in the same direction, the scalar output is naturally the magnitude of a.

[check that it is in fact a legal projection]

The orthogonal projection.

The vector projection can nicely be interpreted from the orthogonal projection.

2.2.10 Notable Vectors

The **zero vector** is a vector composed of zeroes only:

$$\vec{0} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

The **unit vector** is a vector whose length is equal to 1 and is denoted by a *hat*:

$$\hat{i} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \hat{j} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \hat{k} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \hat{v} = \begin{bmatrix} \sqrt{\frac{1}{3}} \\ \sqrt{\frac{1}{3}} \\ \sqrt{\frac{1}{3}} \end{bmatrix}, \ldots$$

The **sparse vector** are vectors where most of its elements are zero:

$$\text{num_zero}(\vec{x}) \gg 0$$

[source]

2.3 Matrices

2.3.1 Matrix Addition

2.3.2 Matrix Subtraction

2.3.3 Matrix Multiplication

2.3.4 Matrix Inverse

2.3.5 Notable Matrices

2.4 Vector Spaces

A **vector space**, also called a **linear space**, is a set (whose elements are called vectors) that can be can be (1) added together and (2) multiplied by scalars. Vector spaces are the basic setting in which linear algebra happens. To comprise a vector space, the operations of vector addition and scalar multiplication must satisfy certain requirements, called vector axioms:

[source] [source]

For a vector space V, for all $u, v, w \in V$, and for all scalars α, $\beta \in F$:

(i) Commutativity of vector addition: $u + v = v + u$

(ii) Associativity of vector addition: $u + (v + w) = (u + v) + w$

(iii) Identity element of vector addition: $\exists\, 0 \in V : v + 0 = v$ where 0 is called the **zero vector**

(iv) Inverse elements of vector addition: $\exists -v \in V : v + (-v) = 0$ where $-v$ is called the **additive inverse** of v

(v) Compatibility of scalar multiplication with field multiplication: $\alpha(\beta v) = (\alpha\beta)v$

(vi) Identity element of scalar multiplication: $1\ v = v$ where 1 is the **multiplicative identity** in F

(vii) Distributivity of scalar multiplication with respect to vector addition: $\alpha(u + v) = \alpha u + \alpha v$

(viii) Distributivity of scalar multiplication with respect to field addition: $(\alpha + \beta)(u + v) = \alpha u + \beta v$

[source]

note: In math, a **field** F is a set on which addition, subtraction, multiplication, and division are defined and behave as the corresponding operations on rational and real numbers do (e.g. R).

[source]

2.4.1 Euclidean Vector Space

Euclidean Vector Space is the quintessential vector space, denoted $\mathbb{R}^n$. Each vector in $\mathbb{R}^n$ has n elements each being a real number:

$$x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \in \mathbb{R}^n, x_i \in \mathbb{R}$$

Proof that $\mathbb{R}^n$ is a vector space:

This demonstrates that we can add any two vectors, and we can multiply vectors by scalars – giving us a well defined vector space.

(i)

(ii)

Euclidean space is the fundamental space of classic geometry. It is used to represent physical space with notions such as distance, length, and angles. For example, $\mathbb{R}^1$ is a line, $\mathbb{R}^2$ is the x-y plane, $\mathbb{R}^3$ is equally familiar. While the space can be hard to visualize for $n > 3$, these concepts generalize nicely to higher dimensions. [source] [source] [source]

More technically, Euclidean vector space is a finite-dimensional inner product space over the real numbers. We will discuss what this means when we visit inner products later on.

2.4.2 Examples of Vector Spaces

2.4.3 Column Space and Row Space

2.4.4 Span, Basis, Dimension

If a vector space V consists of linear combinations of the particular vectors $v_1, v_2, \ldots, v_n$, then we say these vectors **span** the vector space. That is, every vector $w \in V$ can be expressed as a linear combination of $v_1, v_2, \ldots, v_n$:

$$w = a_1 v_1 + a_2 v_2 + \ldots + a_n v_n \text{ for some coefficients } a_i.$$

Since the **spanning set** $v_1, v_2, \ldots, v_n$ can be excessively large, there can be more than one set of coefficients c_i that produce the vector w. [source]

A set of vectors B in a vector space V is a **basis** for V iff its elements are linearly independent and every element of V can be expressed as a linear combination of the vectors in B. [source]

Equivalently, a basis for a vector space is a set of vectors having two properties at once:

(i) It is linearly independent.

(ii) It spans the space.

[source]

Even shorter, a basis is a linearly independent spanning set. [source]

The **dimension** of a vector space V is the **cardinality** (i.e. number of elements) of any basis B for V. All bases of a vector space have equal cardinality, and thus the dimension of a vector space is uniquely defined. We can think of each vector in the basis as a degree of freedom, and thus the dimension of a vector space is the number of degrees of freedom. [source] [source]

For R^3, any basis must contain three vectors, since there are three degrees of freedom (i.e. x-axis, y-axis, and z-axis). And thus R^3 has dimension 3. This same reasoning can be applied to arbitrary R^n.

2.5 Subspace

2.5.1 The Four Fundamental Subspace

2.6 Linear Maps

3 Calculus

3.1 Differential Calculus

3.1.1 Limits [source]

s

$$\lim_{x \to a} f(x) = L$$

iff for every $\epsilon > 0$ there exists a number $\delta > 0$ such that if $0 < |x - a| < \delta$ then $|f(x) - L| < \epsilon$.

ϵ is the parameter for the amount we are away from L. δ is the parameter for the amount that x is away from a. For the limit to be L, then for any ϵ (e.g. a very small one), we can still find a δ (probably a small one), such that if x is close enough to a (off by less than δ), then our function at this range of x will be off by L by at most ϵ.

Simply put, if we want to be close to L by some ϵ, we can always move closer to a to accomplish this. If moving closer to a does not accomplish this (even if we move infinitesimally close), then the limit does not hold.

3.1.2 f continuous at a [source]

We say a function **f is continuous at a** iff

$$\lim_{x \to a} f(x) = f(a)$$

Intuitvely, a function is continuous at a point if there is no break in the graph at the point, i.e. we can draw the graph of the function through that point without lifting our pen. [source]

3.1.3 derivative of f at a (or $f'(a)$ [source]

We say the **derivative of f at a (or $f'(a)$)** is

$$\lim_{h \to 0} \frac{f(a+h) - f(a)}{h}$$

iff this limit exists.

Recall, that the equation for slope of a line is:

$$m = \frac{f(x_2) - f(x_1)}{x_2 - x_1}$$

Since, $h \to 0$, we are in essence finding the slope (or rate of change) of function at a:

$$m = \frac{f(a+h) - f(a)}{(a+h) - a} = \frac{f(a+h) - f(a)}{h}$$

Reminder: For $h \to 0$, we have h approaching 0 from the right ($-$ (e.g. .0001)) and from the left ($+$ (e.g. -.0001)).

Examples where $f'(a)$ does not exist:

i $f'(0) : f(x) = |x|$
One way to intuit is that as you approach from the left the slope is -1 and as you approach from the right the slope is 1. You might think, well the limit as $x \to 0$ of $|x|$ is 0, so the limit does exist and thus it should be differentiable at this point. However, to be differentiable at 0, recall that the slope from the right has to be the same as the slope from the left. You also can't draw a unique tangent line here at $x = 0$. [source]

ii $f'(5) : f(x) = \begin{cases} x^2 & x >= 5 \\ -x & x < 5 \end{cases}$

3.1.4 f differentiable on a [source]

f differentiable at a iff f'(a) exists.

3.1.5 f differentiable on (a, b) [source]

f differentiable on an open interval (a, b) iff it is differentiable at every number in the interval. Example intervals: (a, b), (a, ∞), $(-\infty, b)$, $(-\infty, \infty)$.

We don't consider the endpoints on a closed interval $[a, b]$, since differentiability is only defined for interior points in the vanilla case.

3.1.6 The Extreme Value Theorem [source]

Extreme Value Theorem: If f is continuous on a closed interval $[a, b]$, then f contains an absolute maximum value $f(c)$ and absolute minimum value $f(d)$ for some c, d in the interval.

This doesn't really work for an open interval, since the maximum could be the value as you approach one of the endpoints, therefore, you could always get a larger value by moving closer to the endpoint. If you and try and get cute and make one of the intervals closed, you could come up with a scenario where the other endpoint is the maximum/minimum.

3.1.7 The Mean Value Theorem [source]

Theorem 1. ***The Mean Value Theorem:***

Let f be a function that satisfies the following two hypothesis:

- f is continuous on the closed interval $[a, b]$
- f is differentiable on the open interval (a, b)

Then there is a number c in (a, b) such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

to fill in. □

The mean value theorem connects the average rate of change (right side of the equation) to its derivative. It is saying that there exists a tangent line at some point in the interval that is parallel to the secant line. applications [to fill in]

3.1.8 The Mean Value Theorem [source]

3.2 Integral Calculus

3.3 Multivariable Calculus

3.4 Matrix Calculus

4 Optimization

5 Probability

5.1 Random Variables

5.2 Counting

an event an event is a set

5.3 Chain Rule

For two random *events* A and B, the chain rule claims the following:

$$P(A \cap B) = P(A|B) * P(B) = P(B|A) * P(A)$$

We can think about this problem in two different ways. Way 1: Imagine we are in the space of B (which gives $P(B)$), and now we ask what is the probability of A given that we are in the space of B (which gives $P(B|A)$. Now we might ask why do we multiply the two terms instead of adding them? A mathematical proof can be as followed: Assume A and B are two independent events with $P(A) > .5$ and $P(B) > .5 \Rightarrow P(A \cap B) = P(B|A) * P(A) = P(B) + P(A) > 1$. Contradiction, by the law of total probability.

We can generalize the chain rule to multiple events, $A_1, A_2, ..., A_n$:

$$\begin{aligned} P(A_1 \cap A_2 \cap ... \cap A_n) = & P(A_1|A_2 \cap ... \cap A_n) * P(A_2|A_3 \cap ... \cap A_n) * ... * \\ & P(A_{n-1}|A_n) * (A_n) \end{aligned}$$

More compactly:

$$P(A_1 \cap A_2 \cap ... \cap A_n) = \prod_{i=1}^{n} P\left(A_i \,\middle|\, \bigcap_{j=i+1}^{n} A_j\right)$$

5.4 Baye's Theorem

Baye's Theorem: For two events, A and B (and $P(B) \neq 0$):

$$P(A|B) = \frac{P(B|A) * P(A)}{P(B)}$$

Proof. Using the and rule of probability:

$$\begin{aligned} P(A, B) = P(A|B) * P(B) &= P(B|A) * P(A) \\ P(A|B) * P(B) &= P(B|A) * P(A) \\ P(A|B) &= \frac{P(B|A) * P(A)}{P(B)} \end{aligned}$$

□

While the math is easy to follow, an intuitive proof of Baye's rule is scarcely provided. [add geometric proof] [add intuitive proof]

Baye's theorem (or Baye's rule) relates the **posterior probability** $P(A|B)$ to the **likelihood** $P(B|A)$ using the **prior** $P(A)$ and **marginalization** $P(B)$. The usefulness of this fact cannot be overstated and has made the equation a crown jewel equation of AI. This is because while one conditional can be easy to compute, the other can be tricky or even infeasible. Baye's rule quells this by allowing you to compute one conditional from the other. Here is an example to demonstrate its power: Say you are trying to calculate the probability of someone having lung cancer given that they wake up with a cough. Now this quantity does not seem obvious, neither to estimate or to calculate. However, its subcomponents are. Let:

$$\begin{aligned} A &:= \text{Has lung cancer.} \\ B &:= \text{Wake up with cough.} \\ P(A) &= 0.00094 \text{ [source]} \\ P(B) &= 0.0274 \text{ (estimated at 10/365)} \\ P(B|A) &= 0.57 \text{ [source]} \end{aligned}$$

Using bayes Rule:

$$P(A|B) = \frac{P(B|A) * P(A)}{P(B)}$$
$$P(A|B) = \frac{0.57 * 0.00094}{0.0274}$$
$$P(A|B) = 0.019557$$

6 Machine Learning Problem Formulation

6.1 Supervised Learning

6.2 Unsupervised Learning

6.3 Reinforcement Learning

6.4 Semi-supervised Learning

7 Shallow Learning

8 Deep Learning

9 Resources

9.1 Videos

Vectors — Chapter 1, Essence of linear algebra
Deep Learning State of the Art (2020)

10 After Model Deployment